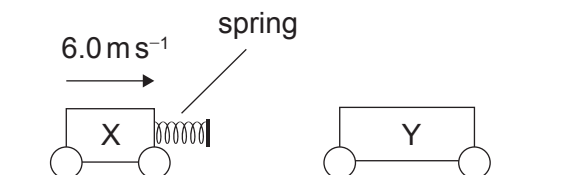
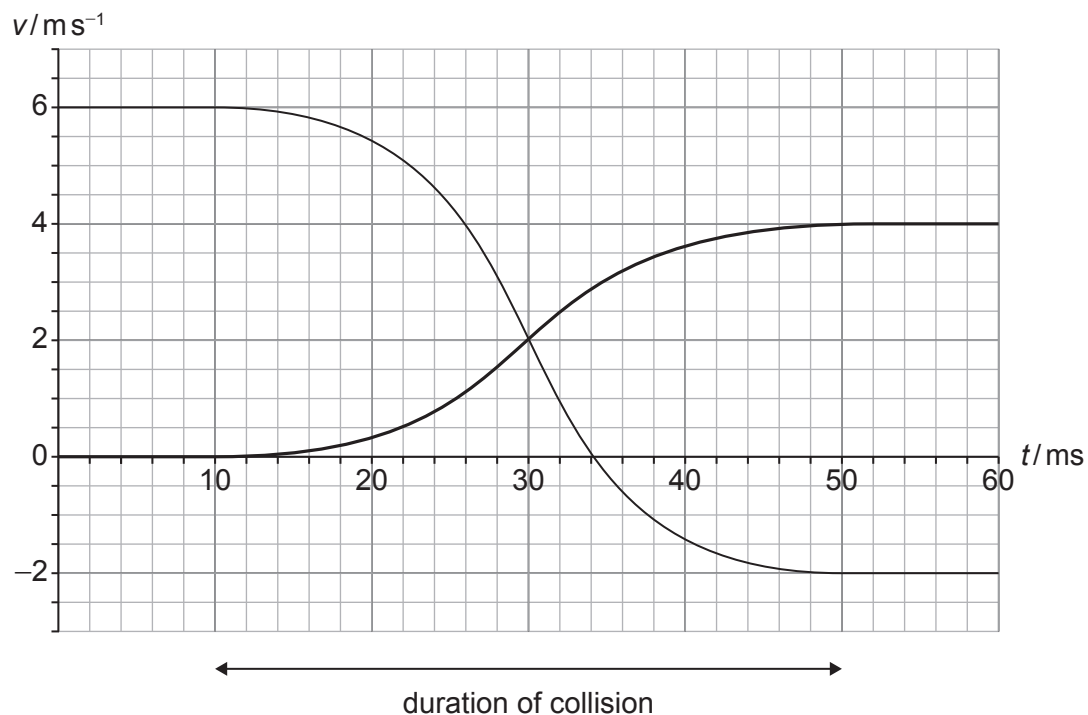


19. A cart X of mass 3.0 kg moving at 6.0 ms^{-1} collides with a stationary cart Y of mass 6.0 kg . A spring is attached to X as shown.



The graph shows the velocities of X and Y before, during and after the collision.



The collision lasted for 40 ms .

- (a) Show that the collision is elastic.

[2]

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(This question continues on the following page)

(b) Determine

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